📄 PROCÉDURE D'INSTALLATION : TP SERVEUR DE MESSAGERIE (DNS + POSTFIX/DOVECOT)

Objectif : Mettre en place une architecture de messagerie complète sur deux machines Debian 12. **Domaine :** `sio2a.home`

MACHINE 1 : LE SERVEUR DNS

- **IP :** `172.17.219.114`
- **Rôle :** Résolution de noms (NS) et enregistrement Mail (MX).

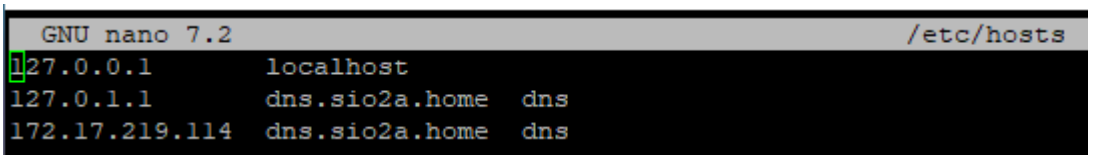
Étape 1 : Identité de la machine

```
# 1. Définir le nom
hostnamectl set-hostname dns.sio2a.home
```

```
# 2. Modifier le fichier hosts
nano /etc/hosts
```

Contenu à mettre dans `/etc/hosts` :

```
127.0.0.1       localhost
127.0.1.1       dns.sio2a.home  dns
172.17.219.114  dns.sio2a.home  dns
```



Étape 2 : Installation et fichiers BIND9

```
apt update
apt install bind9 -y
```

A. Déclarer la zone (`named.conf.local`)

```
nano /etc/bind/named.conf.local
```

Ajoutez ceci à la fin :

```
zone "sio2a.home" {
    type master;
    file "/etc/bind/db.sio2a.home";
};
```

```
  GNU nano 7.2                                  /etc/bind/named.conf.local
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "sio2a.home" {
    type master;
    file "/etc/bind/db.sio2a.home";
};
```

B. Créer la base de données DNS (`db.sio2a.home`)

```
cp /etc/bind/db.empty /etc/bind/db.sio2a.home
nano /etc/bind/db.sio2a.home
```

Copiez exactement ceci (Attention aux points !) :

```
$TTL    86400
@       IN      SOA     dns.sio2a.home. root.sio2a.home. (
                              1         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                          86400 )       ; Negative Cache TTL

@       IN      NS      dns.sio2a.home.

; --- Correspondance Noms -> IP ---
dns     IN      A       172.17.219.114
mail    IN      A       172.17.219.185
@       IN      A       172.17.219.185
```

```
; --- Le plus important : MX (Mail Exchange) ---
@       IN      MX      10      mail.sio2a.home.
```

```
  GNU nano 7.2                                  /etc/bind/db.sio2a.home
; BIND reverse data file for empty rfc1918 zone
;
; DO NOT EDIT THIS FILE - it is used for multiple zones.
; Instead, copy it, edit named.conf, and use that copy.
;
$TTL    86400
@       IN      SOA     dns.sio2a.home.  root.sio2a.home. (
                              1         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                          86400 )       ; Negative Cache TTL
;
@       IN      NS      dns.sio2a.home.

dns     IN      A       172.17.219.114
mail    IN      A       172.17.219.185
@       IN      A       172.17.219.185

@       IN      MX      10      mail.sio2a.home.
```

Étape 3 : Vérification et Démarrage

```
# Vérifie la syntaxe des fichiers
named-checkconf -z
```

```
# Si pas d'erreur, on redémarre
systemctl restart bind9
```

MACHINE 2 : LE SERVEUR MAIL

- **IP :** `172.17.219.185`
- **Rôle :** SMTP (Postfix), POP/IMAP (Dovecot), Webmail (Rainloop).

Étape 1 : Identité et Liaison DNS

```
# 1. Nom
hostnamectl set-hostname mail.sio2a.home
```

```
# 2. Hosts
nano /etc/hosts
```

Contenu `/etc/hosts` :

```
127.0.0.1       localhost
127.0.1.1       mail.sio2a.home  mail
172.17.219.185  mail.sio2a.home  mail
```

```
  GNU nano 7.2                                                  /etc/hosts
127.0.0.1       localhost
127.0.1.1       mail.sio2a.home mail
172.17.219.185  mail.sio2a.home mail
```

3. Configuration DNS (`/etc/resolv.conf`)

```
nano /etc/resolv.conf
```

```
domain sio2a.home
search sio2a.home
nameserver 172.17.219.114
```

```
  GNU nano 7.2                                              /etc/resolv.conf
domain sio.local
search sio.local
nameserver 172.17.219.114
```

(Faites un `ping dns.sio2a.home` pour vérifier que ça marche).

```
root@mail:/# ping dns.sio2a.home
PING dns.sio2a.home (172.17.219.114) 56(84) bytes of data.
64 bytes from 172.17.219.114 (172.17.219.114): icmp_seq=1 ttl=64 time=0.383 ms
64 bytes from 172.17.219.114 (172.17.219.114): icmp_seq=2 ttl=64 time=0.478 ms
64 bytes from 172.17.219.114 (172.17.219.114): icmp_seq=3 ttl=64 time=0.480 ms
64 bytes from 172.17.219.114 (172.17.219.114): icmp_seq=4 ttl=64 time=0.588 ms
^C
--- dns.sio2a.home ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.383/0.482/0.588/0.072 ms
```

Étape 2 : Installation Postfix (SMTP)

```
apt install postfix -y
```

- **Type :** Site Internet
- **Nom de courrier :** `sio2a.home`

Configuration (`main.cf`) :

```
nano /etc/postfix/main.cf
```

Modifiez les lignes suivantes :

```
myhostname = mail.sio2a.home
mydestination = mail.sio2a.home, localhost.sio2a.home, localhost,
sio2a.home
inet_interfaces = all           # Important pour écouter l'extérieur
home_mailbox = MailDir/         # Dossier personnel
```

```
  GNU nano 7.2                                  /etc/postfix/main.cf
# See /usr/share/postfix/main.cf.dist for a commented, more complete version


# Debian specific:  Specifying a file name will cause the first
# line of that file to be used as the name.  The Debian default
# is /etc/mailname.
#myorigin = /etc/mailname

smtpd_banner = $myhostname ESMTP $mail_name (Debian/GNU)
biff = no

# appending .domain is the MUA's job.
append_dot_mydomain = no

# Uncomment the next line to generate "delayed mail" warnings
#delay_warning_time = 4h

readme_directory = no

# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 3.6 on
# fresh installs.
compatibility_level = 3.6



# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = mail.sio2a.home
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
myorigin = /etc/mailname
mydestination = mail.sio2a.home, localhost.sio2a.home, localhost, sio2a.home
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
home_mailbox = MailDir/
```

Utilisateurs :

```
mkdir -p /etc/skel/MailDir      # Pour que les futurs users aient le
dossier
adduser asterix                 # Mettre mot de passe "12345"
adduser obelix                  # Mettre mot de passe "12345"
systemctl restart postfix
```

Étape 3 : Installation Dovecot (POP/IMAP)

```
apt install dovecot-common dovecot-pop3d dovecot-imapd -y
```

A. Autoriser les mots de passe (`10-auth.conf`)

```
nano /etc/dovecot/conf.d/10-auth.conf
```

- Mettre : `disable_plaintext_auth = no`
- Vérifier : `auth_mechanisms = plain login`

```
  GNU nano 7.2                          /etc/dovecot/conf.d/10-auth.conf

# Host name to use in GSSAPI principal names. The default is to use the
# name returned by gethostname(). Use "$ALL" (with quotes) to allow all keytab
# entries.
#auth_gssapi_hostname =

# Kerberos keytab to use for the GSSAPI mechanism. Will use the system
# default (usually /etc/krb5.keytab) if not specified. You may need to change
# the auth service to run as root to be able to read this file.
#auth_krb5_keytab =

# Do NTLM and GSS-SPNEGO authentication using Samba's winbind daemon and
# ntlm_auth helper. <doc/wiki/Authentication/Mechanisms/Winbind.txt>
#auth_use_winbind = no

# Path for Samba's ntlm_auth helper binary.
#auth_winbind_helper_path = /usr/bin/ntlm_auth

# Time to delay before replying to failed authentications.
#auth_failure_delay = 2 secs

# Require a valid SSL client certificate or the authentication fails.
#auth_ssl_require_client_cert = no

# Take the username from client's SSL certificate, using
# X509_NAME_get_text_by_NID() which returns the subject's DN's
# CommonName.
#auth_ssl_username_from_cert = no

# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain login
disable_plaintext_auth = no
```

B. Chemin des mails (`10-mail.conf`)

```
nano /etc/dovecot/conf.d/10-mail.conf
```

- Mettre : `mail_location = maildir:/home/%u/MailDir`

```
  GNU nano 7.2                          /etc/dovecot/conf.d/10-mail.conf
##
## Mailbox locations and namespaces
##

# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
#   %u - username
#   %n - user part in user@domain, same as %u if there's no domain
#   %d - domain part in user@domain, empty if there's no domain
#   %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
#   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%ln/%n:INDEX=/var/indexes/%d/%ln/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:/home/%u/MailDir
```

C. Configuration POP3 (`20-pop3.conf`)

```
nano /etc/dovecot/conf.d/20-pop3.conf
```

- Chercher la ligne `pop3_uidl_format`.
- Enlever le # devant.
- Vérifier qu'elle est bien : `pop3_uidl_format = %08Xu%08Xv`

```
  GNU nano 7.2                                  /etc/dovecot/conf.d/20-pop3.conf
##
## POP3 specific settings
##

# Don't try to set mails non-recent or seen with POP3 sessions. This is
# mostly intended to reduce disk I/O. With maildir it doesn't move files
# from new/ to cur/, with mbox it doesn't write Status-header.
#pop3_no_flag_updates = no

# Support LAST command which exists in old POP3 specs, but has been removed
# from new ones. Some clients still wish to use this though. Enabling this
# makes RSET command clear all \Seen flags from messages.
#pop3_enable_last = no

# If mail has X-UIDL header, use it as the mail's UIDL.
#pop3_reuse_xuidl = no

# Allow only one POP3 session to run simultaneously for the same user.
#pop3_lock_session = no

# POP3 requires message sizes to be listed as if they had CR+LF linefeeds.
# Many POP3 servers violate this by returning the sizes with LF linefeeds,
# because it's faster to get. When this setting is enabled, Dovecot still
# tries to do the right thing first, but if that requires opening the
# message, it fallbacks to the easier (but incorrect) size.
#pop3_fast_size_lookups = no

# POP3 UIDL (unique mail identifier) format to use. You can use following
# variables, along with the variable modifiers described in
# doc/wiki/Variables.txt (e.g. %Uf for the filename in uppercase)
#
#  %v - Mailbox's IMAP UIDVALIDITY
#  %u - Mail's IMAP UID
#  %m - MD5 sum of the mailbox headers in hex (mbox only)
#  %f - filename (maildir only)
#  %g - Mail's GUID
#
# If you want UIDL compatibility with other POP3 servers, use:
#  UW's ipop3d          : %08Xv%08Xu
#  Courier              : %f or %v-%u (both might be used simultaneously)
#  Cyrus (<= 2.1.3)     : %u
#  Cyrus (>= 2.1.4)     : %v.%u
#  Dovecot v0.99.x      : %v.%u
#  tpop3d               : %Mf
#
# Note that Outlook 2003 seems to have problems with %v.%u format which was
# Dovecot's default, so if you're building a new server it would be a good
# idea to change this. %08Xu%08Xv should be pretty fail-safe.
#
pop3_uidl_format = %08Xu%08Xv
```

D. Redémarrage

```
systemctl restart dovecot
```

Étape 4 : Installation Rainloop (Webmail)

```
# 1. Installer les dépendances
apt update
apt install apache2 php libapache2-mod-php php-curl php-xml curl unzip
-y
```

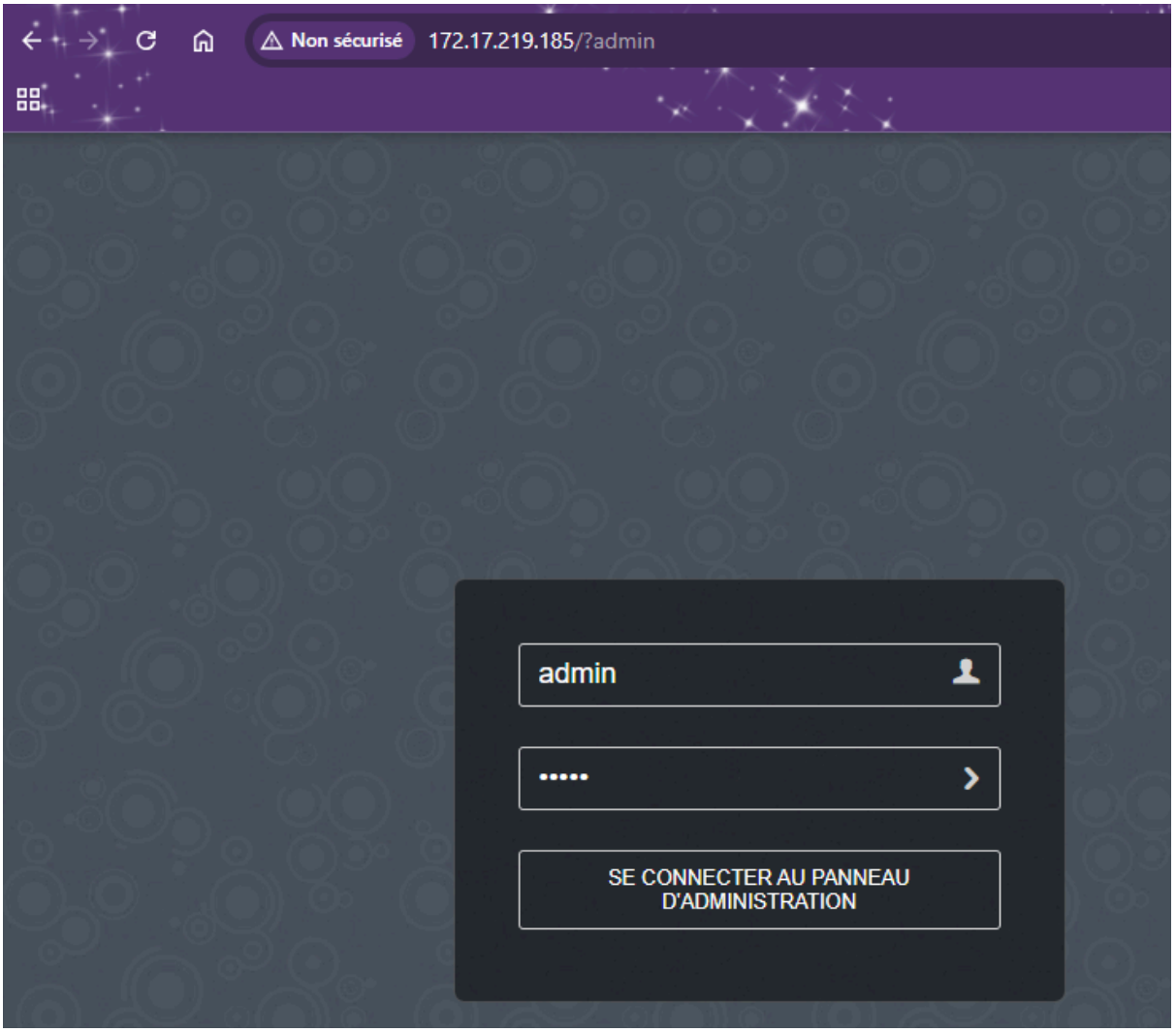
```
# 2. Installer Rainloop (Manuel)
apt update apt install curl unzip php-curl php-xml -y
cd /var/www/html/
rm -rf *
curl -sL https://repository.rainloop.net/installer.php | php
```

```
# 3. Réparer les droits (CRUCIAL)
chown -R www-data:www-data /var/www/html/
find /var/www/html -type d -exec chmod 755 {} \;
find /var/www/html -type f -exec chmod 644 {} \;
```

Étape 5 : Configuration de l'Admin Rainloop (Sur la machine MAIL)

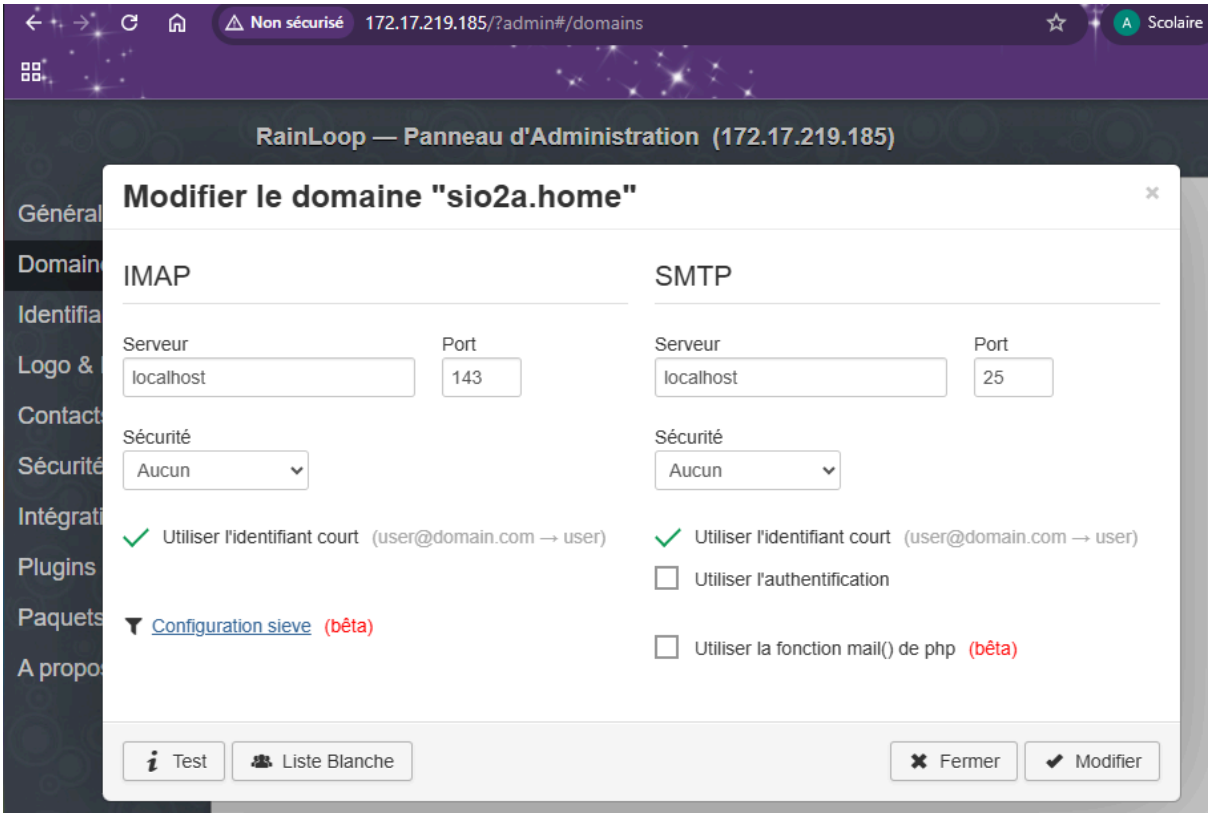
Rainloop est installé mais il ne connaît pas encore votre domaine.

1. Ouvrez un navigateur (depuis votre PC ou une machine du réseau).
2. Accédez à l'URL d'administration : `http://172.17.219.185/?admin`
3. Connectez-vous :
    - **Login :** `admin`
    - **Password :** `12345`



Ajout du domaine :

1. Allez dans l'onglet **Domains**.
2. Cliquez sur le bouton **Add Domain**.
3. Remplissez comme suit :
   - **Name :** `sio2a.home`
   - **IMAP Server :** `localhost` | **Port :** `143`
   - **SMTP Server :** `localhost` | **Port :** `25`
   - **Secure :** `None` (pour les deux).
   - ⚠️ **IMPORTANT :** Cochez la case **"Use short login"** (Utiliser identifiant court).
4. Cliquez sur le bouton **Test**.
   - Si les boutons deviennent **VERTS**, tout est parfait.
5. Cliquez sur **Add** pour valider.



Étape 6 : Les Tests Techniques (Validation du TP)

Faites ces tests **sur la machine MAIL** (172.17.219.185) pour prouver que les services tournent.

Test A : Vérification DNS (Le serveur sait-il où envoyer ?)

```
dig sio2a.home MX
```

- **Résultat attendu :** Une section `ANSWER SECTION` qui affiche `IN MX 10 mail.sio2a.home.`

```
root@mail:/# dig sio2a.home MX

; <<>> DiG 9.18.24-1-Debian <<>> sio2a.home MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62759
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 4d69f58adacd1bdf010000006936ab7790ec3b1562a626cd (good)
;; QUESTION SECTION:
;sio2a.home.                    IN      MX

;; ANSWER SECTION:
sio2a.home.             86400   IN      MX      10 mail.sio2a.home.

;; ADDITIONAL SECTION:
mail.sio2a.home.        86400   IN      A       172.17.219.185

;; Query time: 4 msec
;; SERVER: 172.17.219.114#53(172.17.219.114) (UDP)
;; WHEN: Mon Dec 08 11:41:59 CET 2025
;; MSG SIZE  rcvd: 104

root@mail:/#
```

Test B : Envoi manuel SMTP (Telnet Port 25)

Simulation d'un envoi de mail par ligne de commande (Asterix écrit à Obelix).

```
telnet localhost 25
```

Tapez les commandes suivantes (une par une, validez avec Entrée) :

```
EHLO mail.sio2a.home
MAIL FROM: <obelix@sio2a.home>
RCPT TO: <asterix@sio2a.home>
DATA
Subject: Test Validation TP
Ceci est le mail de validation final.
.
QUIT
```

- **Résultat attendu :** Après le point final, le serveur doit répondre `250 2.0.0 Ok: queued as…`

```
root@mail:/# telnet localhost 25
Trying ::1...
Connected to localhost.
Escape character is '^]'.
220 mail.sio2a.home ESMTP Postfix (Debian/GNU)
EHLO mail.sio2a.home
250-mail.sio2a.home
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
MAIL FROM: <obelix@sio2a.home>
250 2.1.0 Ok
RCPT TO: <asterix@sio2a.home>
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
Subject: Test Validation TP
Ceci est le mail de validation final.
.
250 2.0.0 Ok: queued as 01D9C6D7
```

Test C : Vérification POP3 (Telnet Port 110)

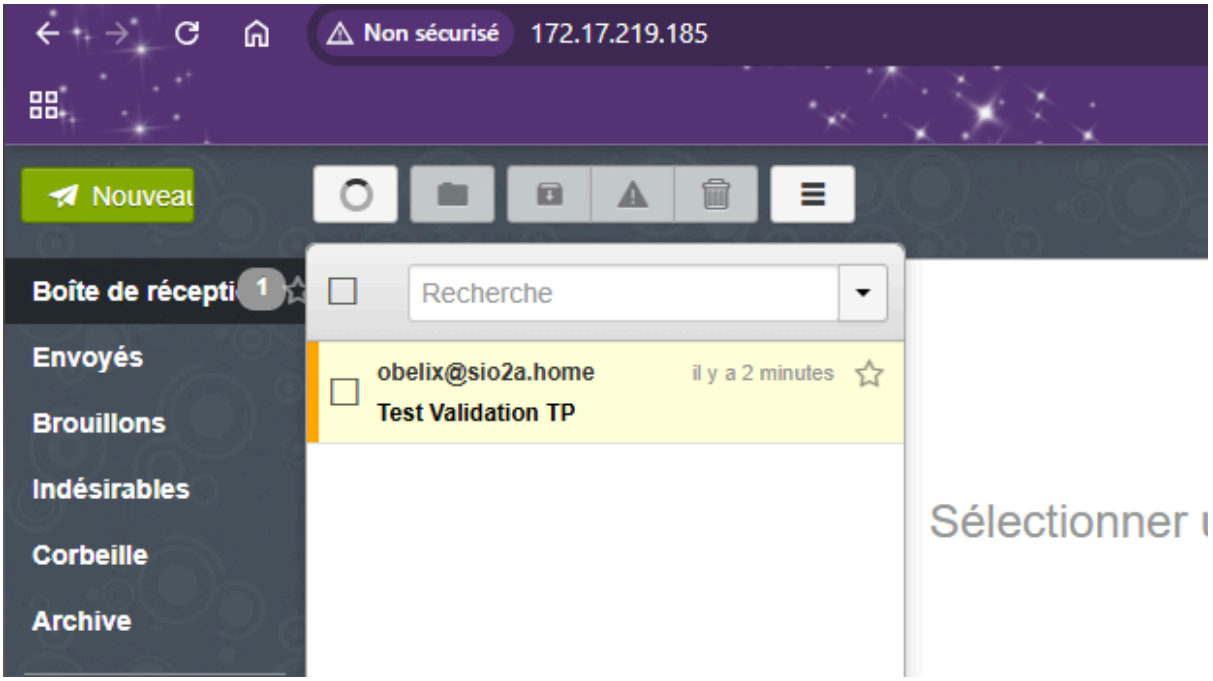
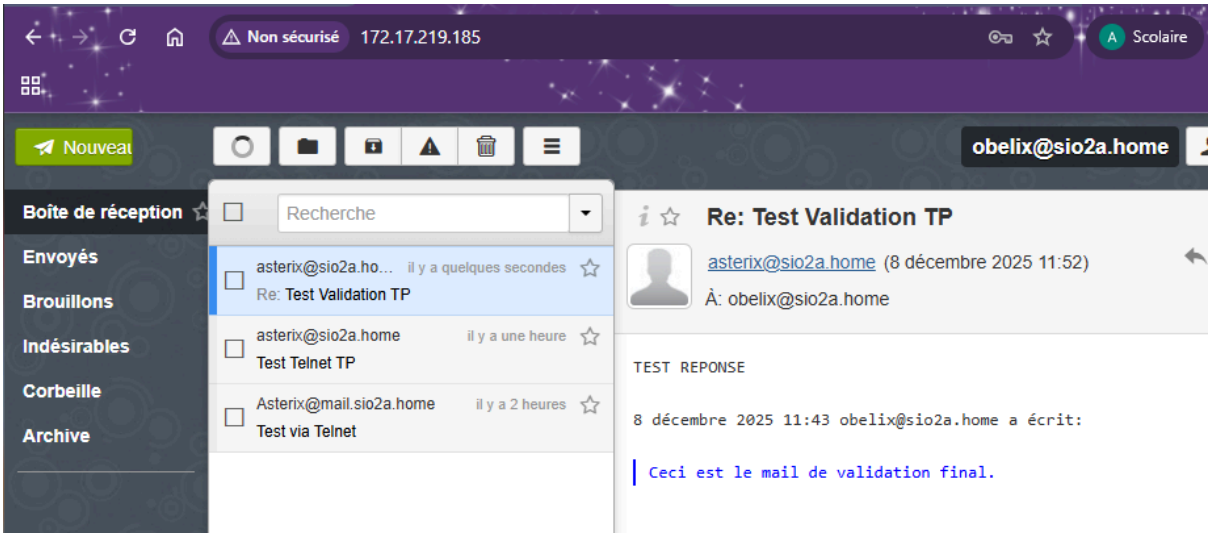
Vérification que Dovecot écoute et que la config UIDL est bonne.

```
telnet localhost 110
```

- **Résultat attendu :** `+OK Dovecot ready.`
- Tapez : `USER obelix`
- Tapez : `PASS 12345`
- **Résultat :** `+OK Logged in.`
- Tapez : `QUIT`

```
root@mail:/# telnet localhost 110
Trying ::1...
Connected to localhost.
Escape character is '^]'.
+OK Dovecot (Debian) ready.
USER obelix
+OK
PASS 12345
+OK Logged in.
```

Étape 7 : Test Utilisateur (Webmail)

C'est la preuve finale que tout fonctionne pour l'utilisateur.

1. Ouvrez votre navigateur sur : `http://172.17.219.185`
2. Connectez-vous :
   - **Utilisateur :** `asterix@sio2a.home`
   - **Mot de passe :** `12345`
3. **Vérification :** *
   - Vous devriez voir dans la boîte de réception le mail **"Test Validation TP"** que vous avez envoyé à l'étape 6B.
   - Essayez de **répondre** à Asterix depuis l'interface.